

14.6 Recommender Systems

Use text (and other signals) to rank items for a user.

1. Suggestions, not a class label

A **recommender** ranks items a user has not seen yet. On the web that is a movie, a product, or "related articles." The NLP angle is that many items **are** text, or have text attached (title, body, reviews).

Two big families:

Collaborative filtering. Recommend what people **like you** liked. Netflix at scale is this idea: your history plus similar users. You need interactions (watches, clicks, ratings). You do not need to understand the plot.

Content-based. Recommend items whose **stuff** looks like items you already used. For news, the stuff is the article text. "Related articles" on a newspaper site is the usual example.

"This book can be for non-black people in Canada who understand that there is a struggle for black liberation in this country — whether they use that language or not. It's for those readers who see demonstrations on the street that are explicitly about blackness by black people and are curious and who want to understand. *The Skin We're In* exists so those who are not black can better articulate the struggle that they see or hear about — so that they can better understand and contextualize it.

"I had the great privilege and honour of taking the time to find all of these things and put them together for the book."

- [7 works of Canadian nonfiction to read for Black History Month 2020](#)

Desmond Cole's comments have been edited for length and clarity.

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A production system often mixes both. This note is about the content side, because that is the topic-model and retrieval machinery from this module.

2. Topic models as a content engine

Take LDA or NMF from Weeks 13–14. Each document gets a topic mix (θ). Two articles with close (θ) (cosine, or a small Euclidean distance) are "about the same things."

That is a content recommender:

1. represent each document (topics, TF-IDF, or an embedding)
2. at read time, score neighbors of the current document
3. show the top few that are not the same URL

You can swap the representation. Topics are interpretable ("we showed this because both are *elections* + *courts*"). Dense embeddings often rank better and explain worse.

Collaborative filtering fails on a brand-new article with no clicks. Content-based does not. That **cold start** is why newsrooms keep a text similarity fallback.

3. How you know it worked

Offline accuracy on a toy split is not the product metric.

Check	What you measure
Business counters	Click-through, dwell, purchase if there is one, engagement
A/B test	Two rankers, two user groups, same counters
User study	People rate "was this related?" on a sample
Tiny gold set	For a fixed article, a hand-made list of good neighbors; compare overlap
Analytics	Dashboards (Google Analytics or your own) so a drop is visible

Industry systems combine these. A ranker that wins ROUGE or NPMI but loses clicks loses.

4. Pre-processing is part of the ranker

Tokenization, lowercasing, and what you strip change the neighbors.

Decide the **objective** first. If relatedness should ignore author bylines, strip them. If "same city" matters, keep place names. If boilerplate ("Subscribe to our newsletter") is in every file, it will glue unrelated articles together. That is a cleanup bug, not a model bug.

The same lesson as KPE and summarization: run the cheap filters before you tune (k).

5. A minimal content stack

For a homework-sized news recommender:

1. Clean boilerplate.
2. Fit TF-IDF or a 20-topic NMF/LDA on the archive.
3. For article (i), take the 5 nearest neighbors by cosine.
4. Drop near-duplicates (same title, or cosine (≈ 1)).

5. Measure overlap with a 20-article hand list, then (if you can) clicks.

That is enough to feel the difference between "same topic" and "same story reprinted." Collaborative filtering can wait until you have a user-item matrix. Evaluation of the topic piece itself is note **14.3**.

6. Practice

1. A brand-new op-ed has zero clicks. Why might collaborative filtering recommend nothing useful, and what do you use instead?
2. Two articles share a high (θ) on a "subscribe / cookie banner" topic. What do you fix?
3. Name two metrics you would put on an A/B test for "related articles," and one metric you would **not** treat as a ship decision by itself.

Clicks without dwell can mean bait. Read both numbers before you ship.